Machine Learning

Dr. Shylaja S S

Professor

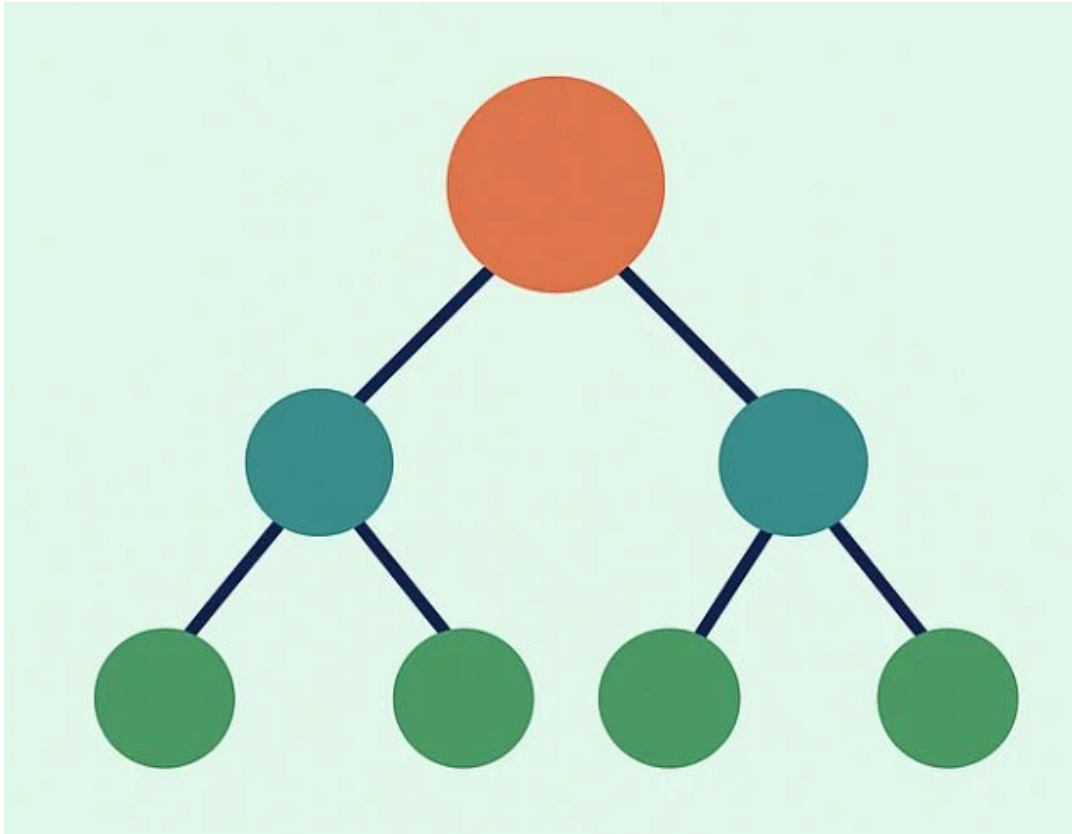
Department of Computer Science & Engineering

Acknowledgement : Dr. Rajanikanth K (Former Principal MSRIT, PhD(IISc), Academic Advisor, PESU)

Prof. Preet Kanwal (Associate Professor, PESU)

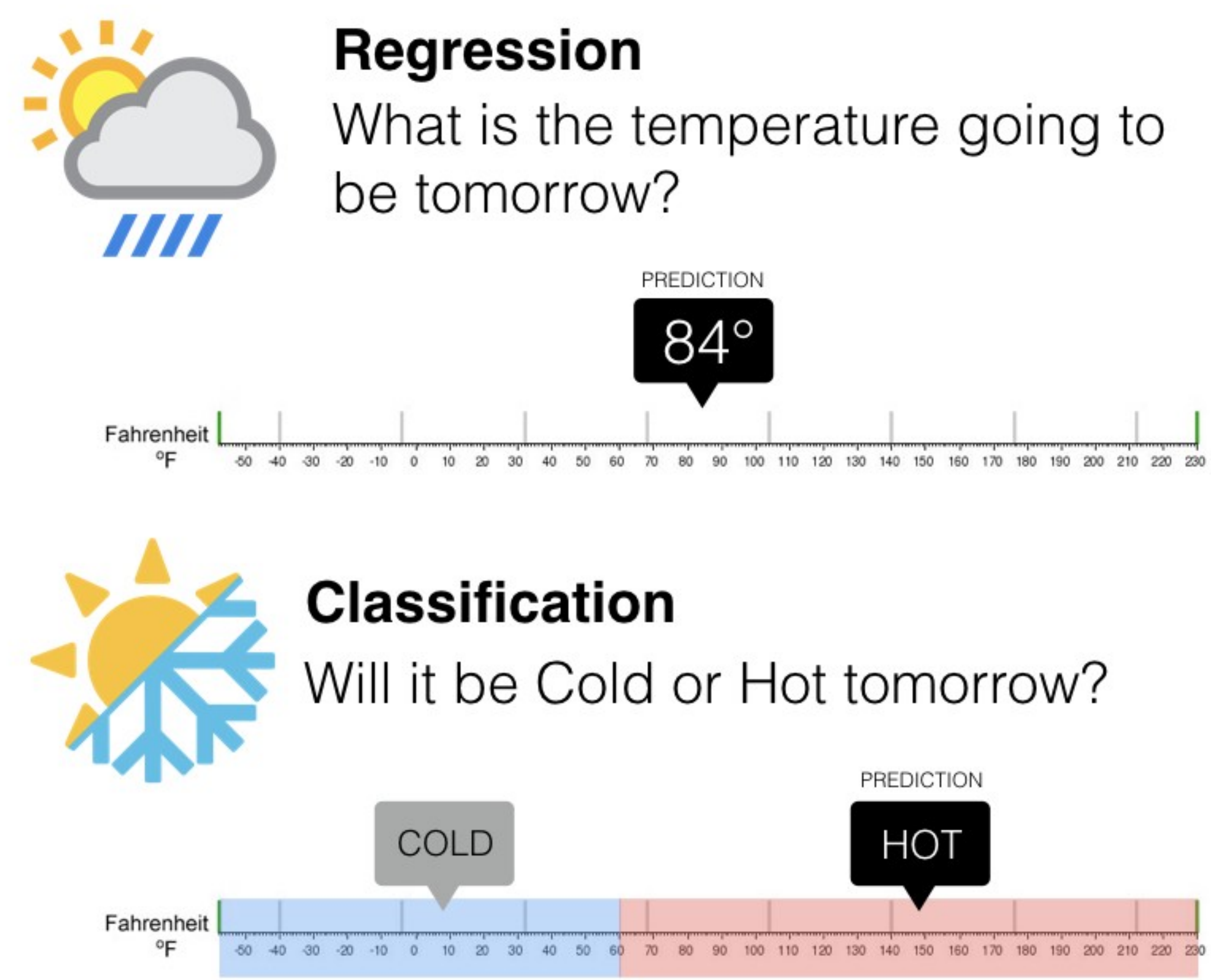
Teaching Assistant (Arya Rajiv Chaloli)

- Decision tree is a flowchart-like diagram to make decisions based on a series of conditions.
- The learned function is represented by a decision tree.
- Decision tree can also be re-represented as **if-then rules** to improve human readability.

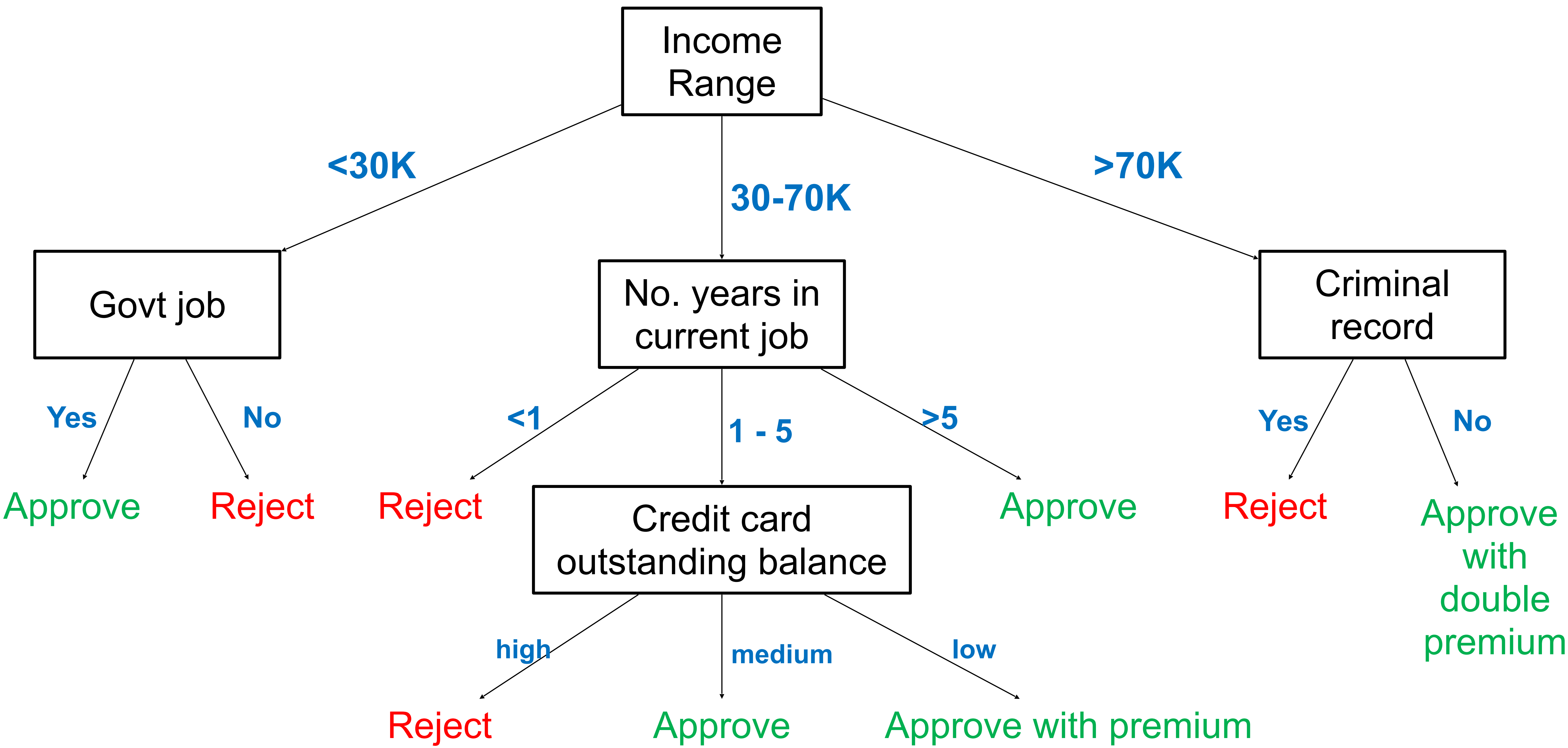


- Decision trees can be used for classification and regression problems

- Decision trees classify instances by sorting them down the tree
from the root to some leaf node.
    - **Node**: Specifies some **attribute** of an instance to be tested
    - **Branch**: Corresponds to one of the possible **possible values** for an attribute



Decision Tree (multiple branches) - Loan Application

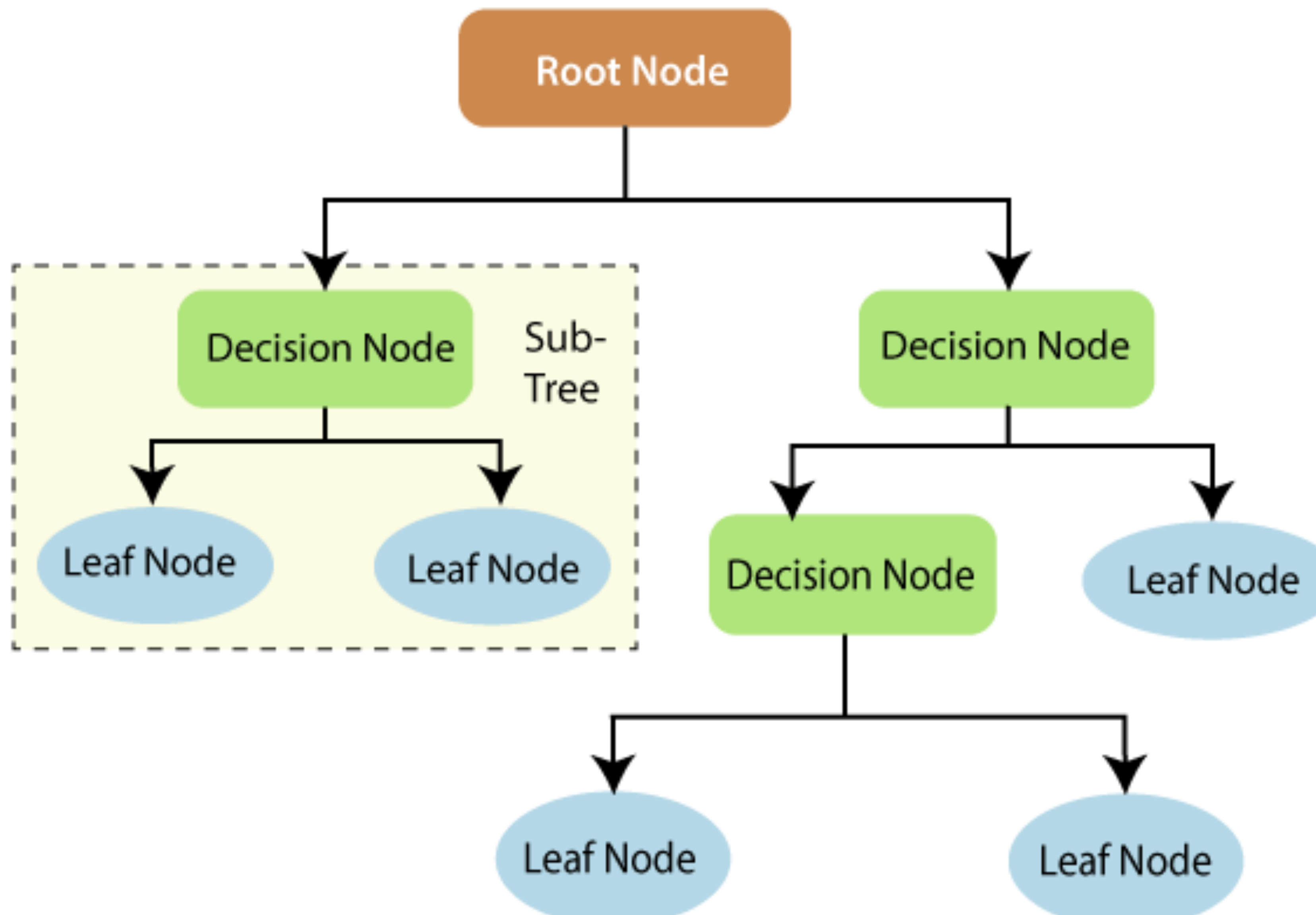


Terminology

Root Node:
It's the topmost node where we start building the tree.

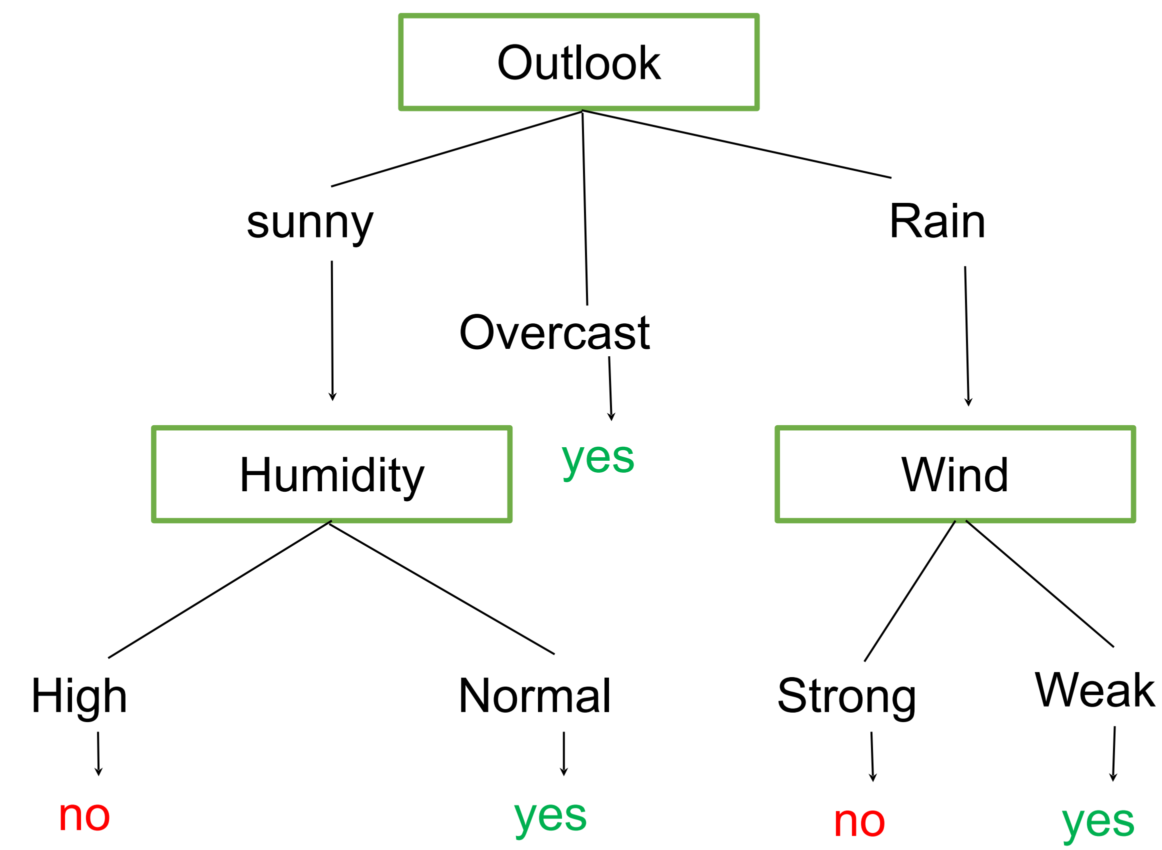
Leaf node:
It's the end of the decision tree and carries the decision.

Decision Node:
Is a node where branching (testing) occurs. Certain attribute is tested and based on the attribute value, a particular branch is taken.



- Every node except leaf node is a decision node.
- Each decision node tests for an attribute.
- Each branch corresponds to an attribute value.
- Each leaf node assigns a classification label.
- Decision trees represent a **disjunction of conjunctions** **(more expressive than FIND-S).**

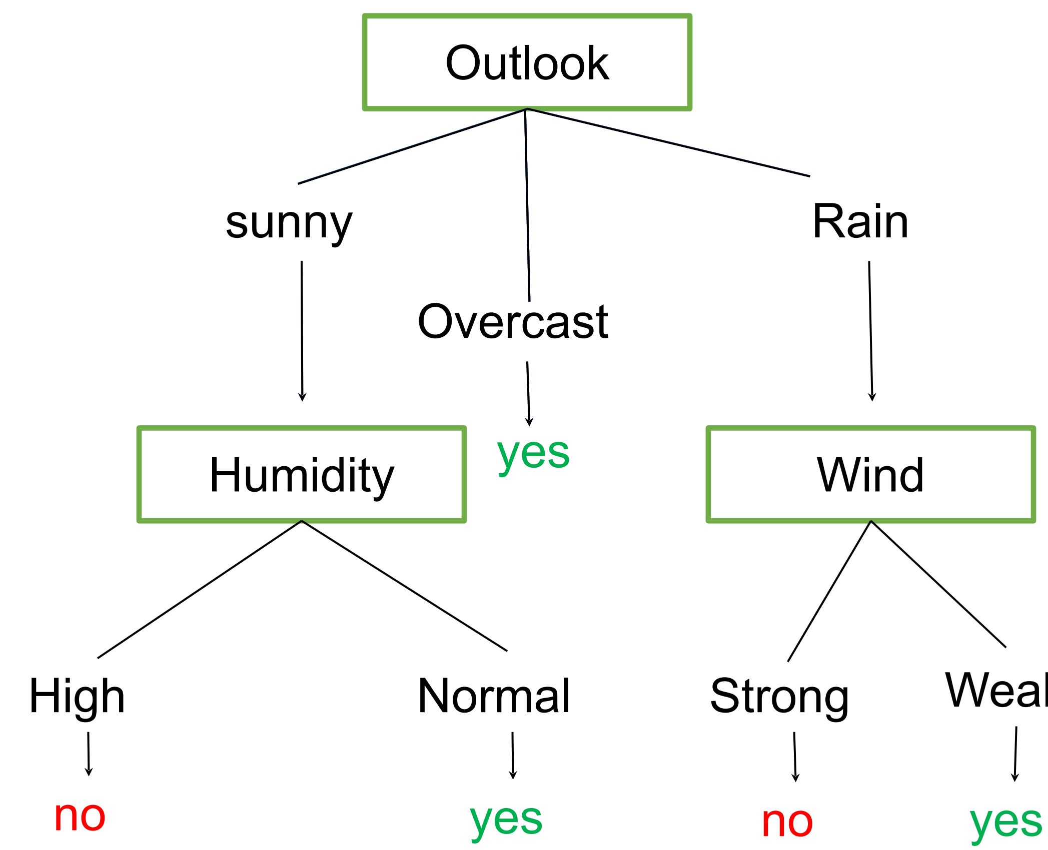
Example decision tree for concept to play Tennis



For example, the decision is yes whenever: (Outlook = Sunny ^ Humidity = Normal) OR (Outlook = Overcast) OR (Outlook = Rain ^ Wind = Weak)

1. Select the **"best" decision attribute** for the next node - **crux of the problem**
2. Assign that attribute as the decision attribute for node.
3. For each value of the attribute, create new descendant of node (new branch is opened).
   - It will depend on the subset of instances that fall under that branch.
   - If no further decision making is required, you will reach the leaf node.
4. Sort training examples to new nodes.
5. If training examples perfectly classified, then STOP, else iterate over new leaf nodes.

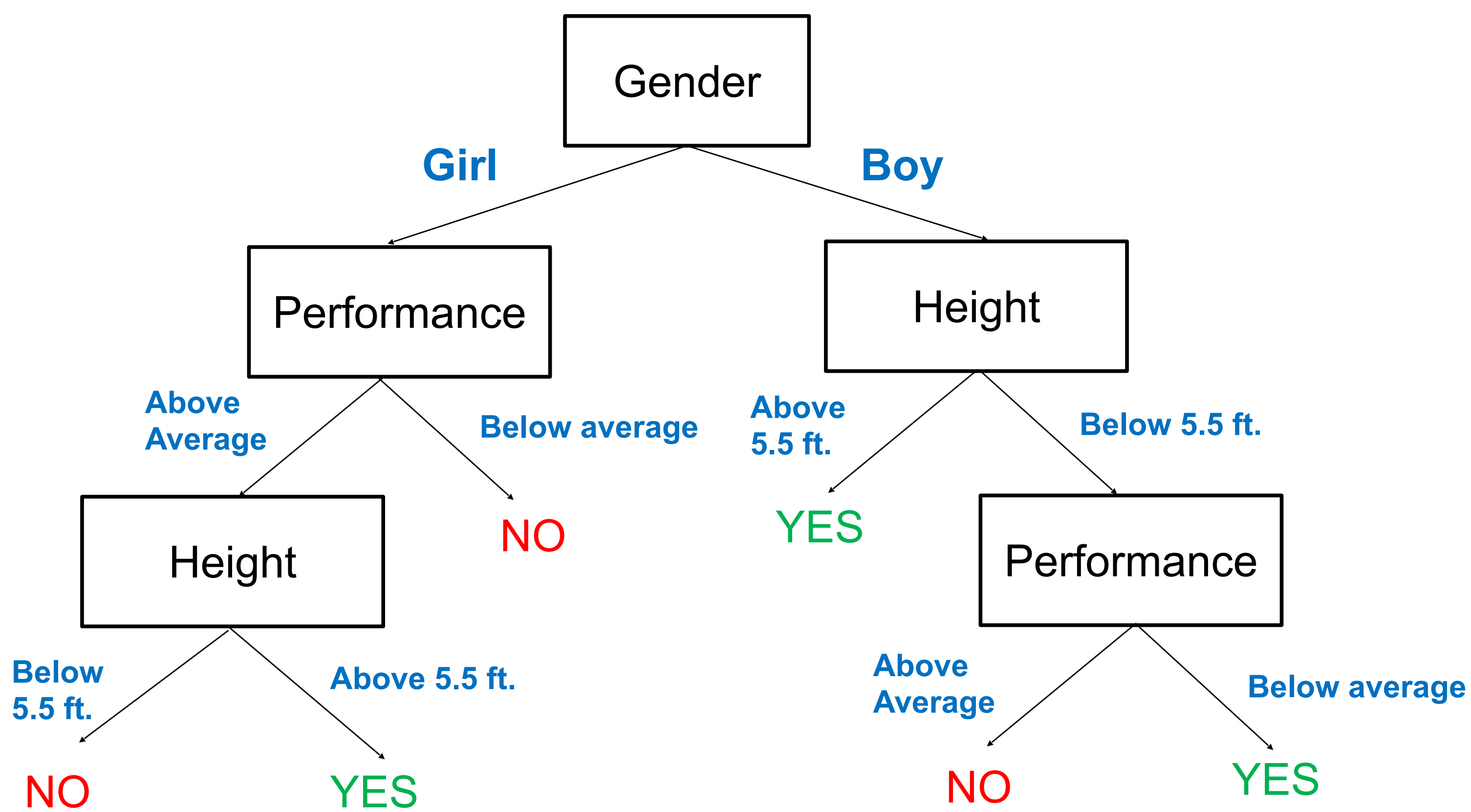
Decision tree for concept to play Tennis

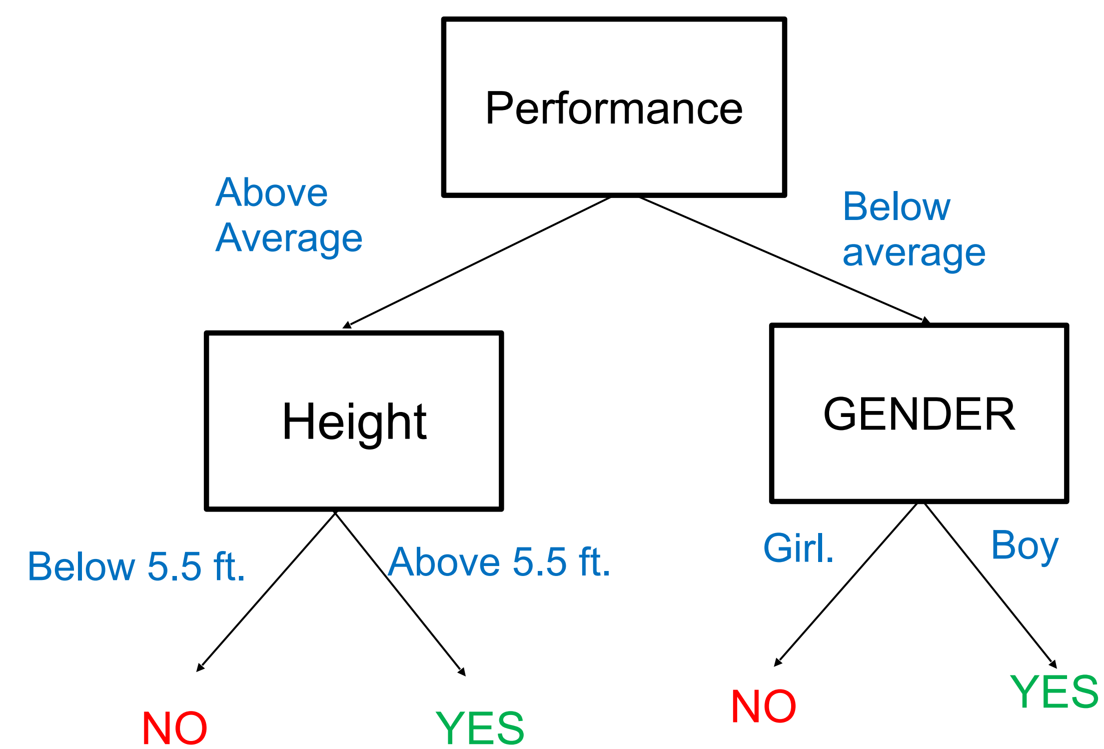


Multiple decision trees for the same task

Decision trees vary based on the algorithm used to select the next **“best” decision attribute** node

Example: To find whether a student plays basketball or not





Which one do we prefer and why?

ID3 Algorithm
(Iterative Dichotomiser 3)

It was developed in 1986 by Ross Quinlan.
The algorithm creates a multiway tree, finding for each node (i.e. in a greedy manner) the categorical feature that will yield the largest *information gain* for the targets.

It can be used for classification
regression.

CART

Classification and Regression Trees or CART for short is a term introduced by Leo Breiman. It can be used for classification or regression predictive modeling problems.

It uses **Gini Impurity** as the heuristic, for classification, and sum of least squares as the heuristic for regression.

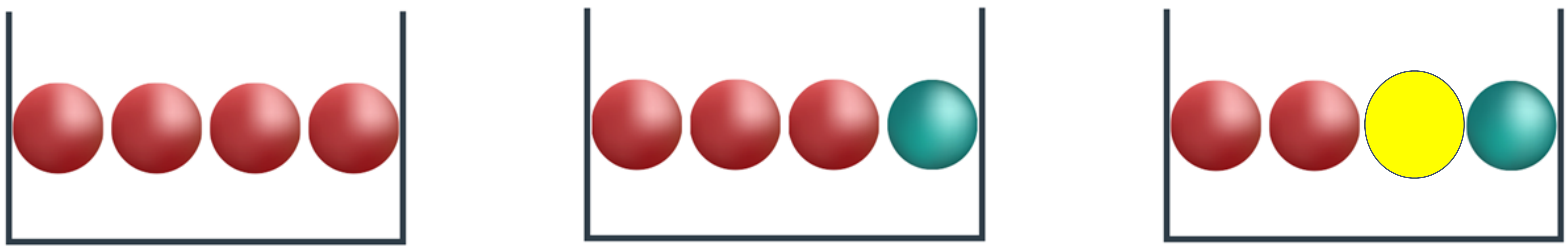
- At a given point, we prefer an attribute as the decision node that partitions the set into subsets which are as homogeneous as possible. Ideally, no more decisions are required!

- Entropy is an indicator of non-homogeneity / uncertainty. It is a measure of disorder (indicator of how messy your data is)
    - **Entropy needs to be reduced.**

- Higher the entropy, the more non-homogeneous / uncertain / disordered the data set is.

- If the attribute A has c possible values, its entropy with respect to a set S is given by:

$$\text{Entropy}(A) = H(A) = -\sum p(x_i) \log_2 (p(x_i)); \; i = 1 \text{ to } c$$



BUCKET 1

LESS ENTROPY

We know for sure that the ball coming out is red.

BUCKET 2

MEDIUM ENTROPY

We know with 75% certainty that the ball is red, and with 25% certainty that it’s green.

BUCKET 3

HIGH ENTROPY

We know with 50% certainty that the ball is red, and 25% certainty with yellow and green each.

- So, we have more knowledge regarding bucket 1 and bucket 1 has the least entropy.

ID3 Algorithm - Information Gain

- Information gain is a measure of the effectiveness of an attribute in classifying the training data.

- **Information Gain = (Entropy before split – Entropy after split)**
    - **Information gain should be as high as possible**

- Information gain is the expected reduction in entropy caused by partitioning the examples according to the attribute.

- **Average information entropy: I(A)** is the weighted average of entropies of individual subsets.

$$I(A) = \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, A) \equiv Entropy(S) - \mathbf{I(A)}$$

where $Values(A)$ is the set of all possible values for attribute A, and S_v is the subset of S for which attribute A has value v (i.e., $S_v = \{s \in S | A(s) = v\}$). Note

Steps to create a decision tree using the ID3 algorithm

1. Compute the **entropy** for data-set **entropy(s)**

1. For each attribute:
   - Calculate entropy for all other values **entropy(a)**
   - Take **average information entropy** for the current attribute
   - Calculate **gain** for the current attribute

3. Pick the **highest gain attribute**

4. **Repeat** until we get the tree we desire.

Construct Decision Tree

Outlook	Temp	Humidity	Windy	Play tennis
Sunny	High	High	Weak	No
Sunny	High	High	Strong	No
Overcast	High	High	Weak	Yes
Rainy	Medium	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Medium	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Medium	Normal	Weak	Yes
Sunny	Medium	Normal	Strong	Yes
Overcast	Medium	High	Strong	Yes
Overcast	High	Normal	Weak	Yes
Rainy	Medium	High	Strong	No

Example 1

Entropy Calculation before Split

Outlook	Temp	Humidity	Windy	Tennis
Sunny	High	High	Weak	No
Sunny	High	High	Strong	No
Overcast	High	High	Weak	Yes
Rainy	Medium	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Medium	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rainy	Medium	Normal	Weak	Yes
Sunny	Medium	Normal	Strong	Yes
Overcast	Medium	High	Strong	Yes
Overcast	High	Normal	Weak	Yes
Rainy	Medium	High	Strong	No

ENTROPY(S) - Calculating for entire Set S

|S| = 14 (training instances provided)

P(play tennis = yes) = p1 = 9/14 = 0.643

P(Play tennis = no) = p2 = 5/14 = 0.357

$$Entropy(S) \equiv \sum_{i=1}^{c} -p_i \log_2 p_i$$

$$= -(p1 \log_2 p1 + p2 \log_2 p2)$$

$$= 0.94$$

Calculating Entropy for a Subset

- We will now see, What happens if we split based on:
  - Outlook
  - Temp
  - Humidity
  - Windy

- First, we will see what will happen If we split based on Outlook.
  - Outlook has 3 values that means there will be three branches (three subsets) = sunny, overcast and rain.
  - We will calculate the entropy for each of these subsets.
  - We take a weighted sum of the entropy's of the three subsets which becomes the "Entropy after the split".
  - We then calculate the gain(Outlook) = Entropy before split - Entropy after split

- We repeat this process for each attribute.

- The attribute with the maximum gain is picked as the decision node.

Calculating Entropy for a Subset

PES UNIVERSITY

Outlook	Temp	Humidity	Windy	Play tennis
Sunny	High	High	Weak	No
Sunny	High	High	Strong	No
Sunny	Medium	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Medium	Normal	Strong	Yes
Overcast	High	High	Weak	Yes
Overcast	Cool	Normal	Strong	Yes
Overcast	Medium	High	Strong	Yes
Overcast	High	Normal	Weak	Yes
Rainy	Medium	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Rainy	Medium	Normal	Weak	Yes
Rainy	Medium	High	Strong	No

Entropy after Split based on Outlook and Gain Calculation

$|S| = 14$, Entropy(S) = 0.94

- Splitting based on Outlook - There will be three subsets S_{sunny} , $S_{overcast}$, S_{rain} such that $|S_{sunny}|= 5$, $|S_{overcast}|= 4$ and $|S_{rain}|= 5$

- Entropy of these subsets:
  - $E(S_{sunny}) = -(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5}) = 0.971$
  - $E(S_{overcast}) = -(1 \log_2 1 + 0 \log_2 0) = 0$
  - $E(S_{rain}) = -(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5}) = 0.971$

- Avg Information of Outlook : weighted avg of the entropies of the subsets:
  $I(Outlook) = \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) = 5/14 * 0.971 + 4/14 * 0 + 5/14 * 0.971$
  $= 0.693$

- Gain(S, outlook) = Entropy(S) - I(outlook) = 0.94 - 0.693 = 0.247

Entropy after Split based on Temp and Gain Calculation

$|S| = 14$, Entropy(S) = 0.94

- Splitting based on Temp - There will be three subsets S_{high} , S_{medium} , S_{cool} such that $|S_{high}|= 4$, $|S_{medium}|= 6$ and $|S_{cool}|= 4$

- Entropy of these subsets:
  - $E(S_{high}) = -(2/4 \log_2 2/4 + 2/4 \log_2 2/4) = 1$
  - $E(S_{medium}) = -(4/6 \log_2 4/6 + 2/6 \log_2 2/6) = 0.918$
  - $E(S_{cool}) = -(3/4 \log_2 3/4 + 1/4 \log_2 1/4) = 0.81125$

- Avg Information of Outlook : weighted avg of the entropies of the subsets:
  $I(Temp) = \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) = 4/14 * 1 + 6/14 * 0.918 + 4/14 * 0.81125$
  $= 0.9109 = 0.911$

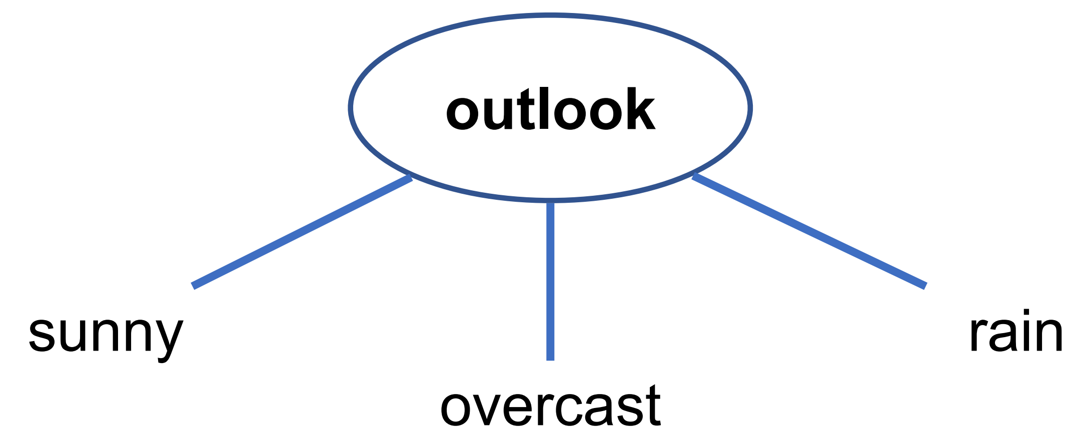
- Gain(S, Temp) = Entropy(S) - I(Temp) = 0.94 - 0.911 = 0.029

Picking an Attribute as Decision Node

- We do a similar calculation to calculate the following:
  - Gain(S, Humidity)
  - G(S, Windy)

- Hence, we can summarize the calculations as:

property	value
Entropy(S)	0.94
G(S, Outlook)	**0.247**
G(S, Temp)	0.029
G(S, Humidity)	0.152
G(S, Windy)	0.048

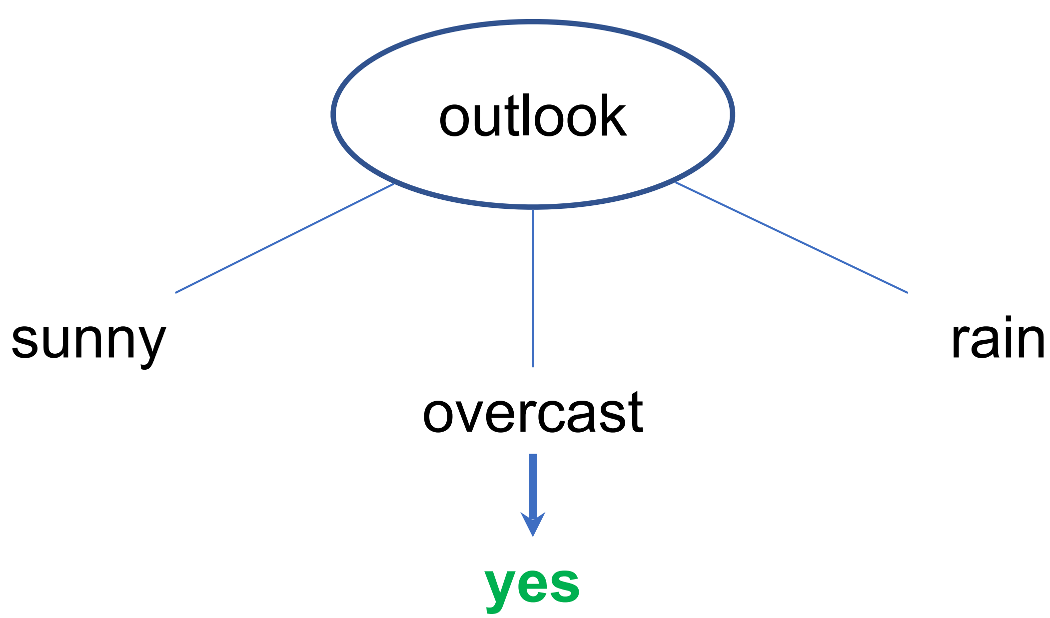


- We pick the **attribute with maximum Gain** as the decision node.

Arriving at a Decision Node - Attribute with highest Gain

- We choose our root node as outlook and split according to its three values.

- Entropy of overcast was 0 which means it is single valued and hence we can come to a conclusion from overcast .

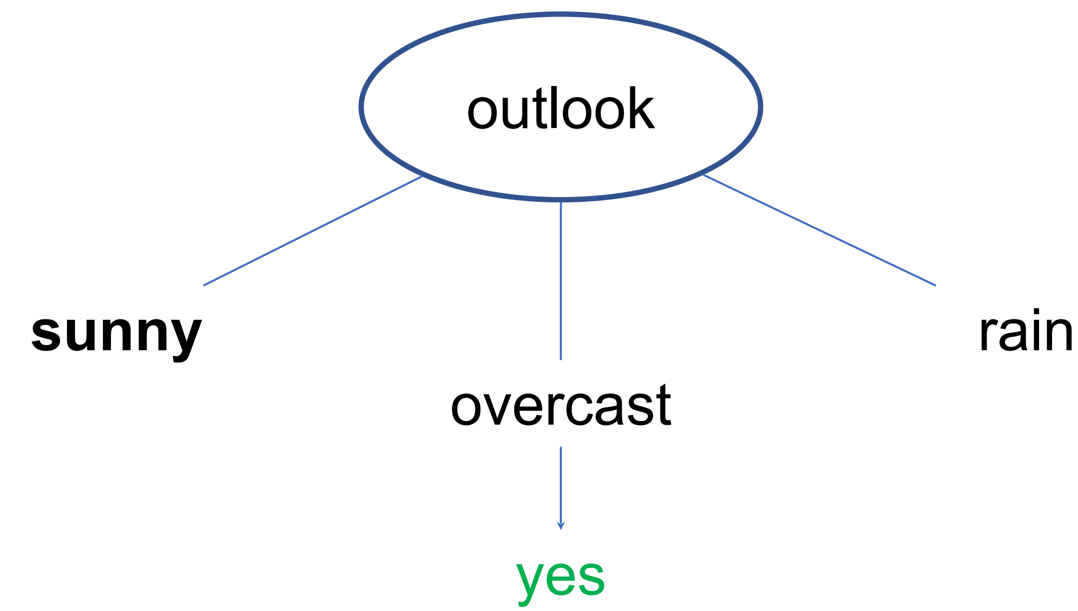


$E(S_{sunny})$	0.971
$E(S_{Overcast})$	0
$E(S_{rain})$	0.971

Repeat the process to pick the descendant node for each branch

With Outlook = Sunny, data would look like this:

Outlook	Temp	Humidity	Windy	Play tennis
Sunny	High	High	Weak	No
Sunny	High	High	Strong	No
Sunny	Medium	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Medium	Normal	Strong	Yes



$E(S_{sunny})$	0.971
$E(S_{Overcast})$	0
$E(S_{rain})$	0.971

Entropy after Split based on Temp and Gain Calculation

$|S_{sunny}| = 5$, Entropy(S_{sunny}) = 0.971

- Splitting based on Temp - There will be three subsets S_{high} , S_{medium} , S_{cool} such that $|S_{high}|= 2$, $|S_{medium}|= 2$ and $|S_{cool}|= 1$

- Entropy of these subsets:
  - $E(S_{high}) = - (0 \log_2 0 + 1 \log_2 1) = 0$
  - $E(S_{medium}) = - (½ \log_2 ½ + ½ \log_2 ½) = 1$
  - $E(S_{cool}) = - (1 \log_2 1 + 0 \log_2 0) = 0$

- Avg Information of Outlook : weighted avg of the entropies of the subsets:
  I(Temp) = $\sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$ = ⅖ * 1 = 0.4

- Gain(S_{sunny}, Temp) = Entropy(S_{sunny}) - I(Temp) = 0.971 - 0.4 = 0.571

Entropy after Split based on Humidity and Gain Calculation

$|S_{sunny}| = 5$, $Entropy(S_{sunny}) = 0.971$

- Splitting based on Humidity - There will be two subsets S_{high} , S_{normal} such that $|S_{high}| = 2$, $|S_{normal}| = 3$

- Entropy of these subsets:
  - $E(S_{high}) = -(0 \ \log_2 0 \ + 1 \ \log_2 1) = 0$
  - $E(S_{normal}) = -(1 \log_2 1 + 0 \log_2 0) = 0$

- Avg Information of Outlook : weighted avg of the entropies of the subsets: I(Humidity) = $\sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) = 0$

- $Gain(S_{sunny}, Humidity) = Entropy(S_{sunny}) - I(Humidity) = 0.971 - 0 = 0.971$

Entropy after Split based on Windy and Gain Calculation

$|S_{sunny}| = 5$, $Entropy(S_{sunny}) = 0.971$

- Splitting based on Humidity - There will be two subsets S_{weak} , S_{strong} such that $|S_{weak}| = 3$, $|S_{strongl}| = 2$

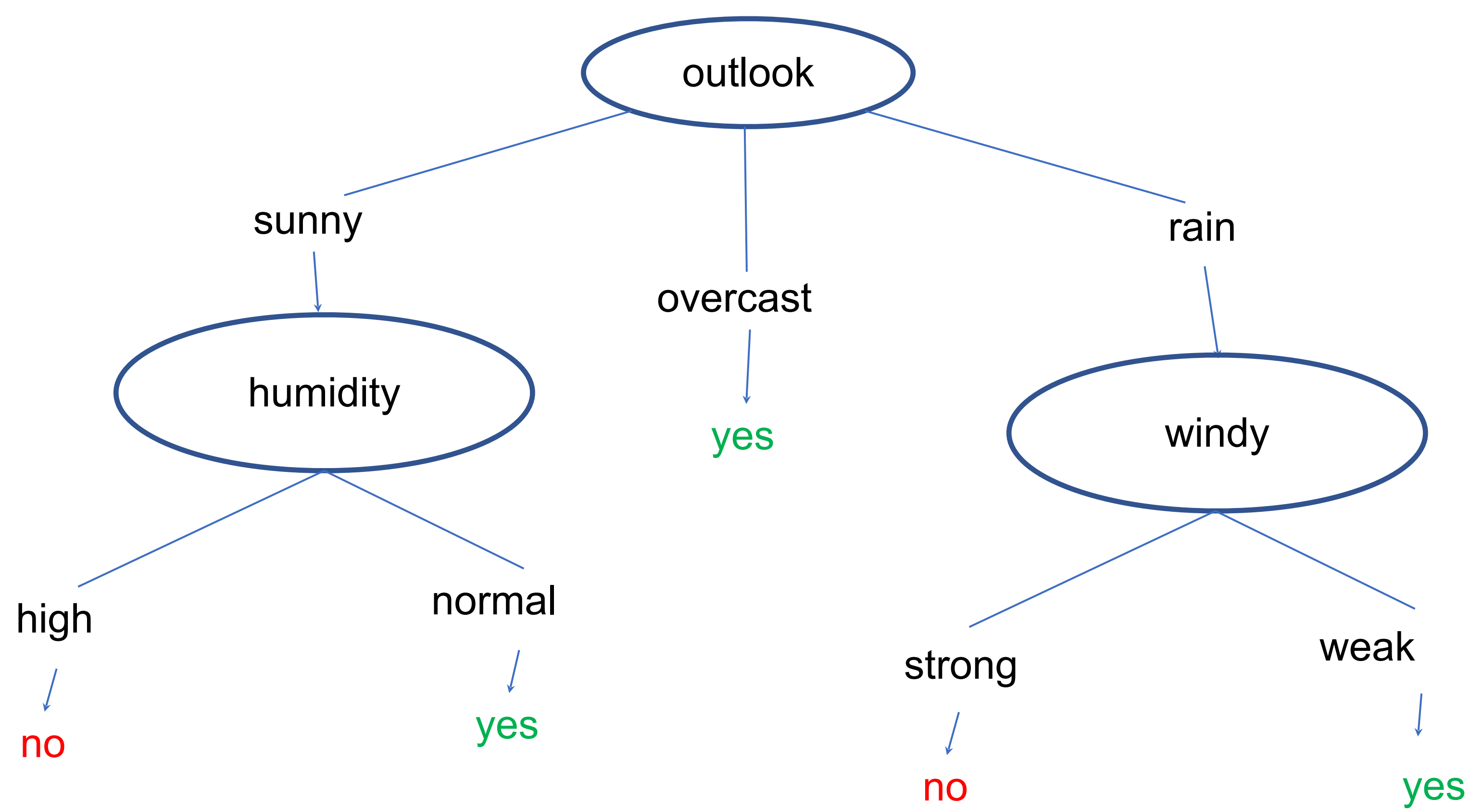
- Entropy of these subsets:
  - $E(S_{weak}) = -(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3}) = 0.918$
  - $E(S_{strong}) = -(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}) = 1$

- Avg Information of Outlook : weighted avg of the entropies of the subsets: $I(Windy) = \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) = 0.951$

- $Gain(S_{sunny}, Windy) = Entropy(S_{sunny}) - I(Windy) = 0.971 - 0.951 = 0.02$

Arriving at a Decision node

- Humidity has the highest gain hence we choose humidity at second level after outlook as sunny
- Remember that both value of humidity (high and normal) had entropy of 0 which means we have reached the leaf node with this.
- Doing the same procedure for the table with outlook=rain we expand the rain leading node to get the following decision tree.



property	value
Entropy(S_{sunny})	0.97
G($S_{sunny,}$temp)	0.571
G(S_{sunny},humidity)	0.971
G(S_{sunny},windy)	0.02

salary	Location	job acceptance
Tier1	MUM	YES
Tier 1	BLR	YES
Tier 2	BLR	NO
Tier 1	HYD	NO
Tier 2	MUM	YES
Tier 1	HYD	NO

Ensure there are no duplicate rows in your data

Example 2

consider the data set of concept of accepting a job

Construct Decision Tree

salary	Location	job acceptance
Tier1	MUM	YES
Tier 1	BLR	YES
Tier 2	BLR	NO
Tier 1	HYD	NO
Tier 2	MUM	YES

Solution

Step 1 : Entropy of the entire set:

$|S| = 5$

P(yes) = 3/5, P(no) = 2/5

Entropy(S) = 0.971

Step 2 : Entropy after Split

a)Based on salary

→ $|S_{Tier1}| = 3$

P(yes) =2/3 , P(no) = 1/3

Entropy(S_{Tier1}) = 0.918

→ $|S_{Tier2}| = 2$

P(yes) = ½ , P(no) = ½

Entropy(S_{Tier2}) = 1

→ I(salary) = 3/5 * 0.918 + 2/5 * 1 = 0.9508

→ Gain(S, salary) = 0.971 - 0.9508=0.0202

Machine Learning

Construct Decision Tree

salary	Location	job acceptance
Tier1	MUM	YES
Tier 1	BLR	YES
Tier 2	BLR	NO
Tier 1	HYD	NO
Tier 2	MUM	YES

Solution

Step 1 : Entropy of the entire set:

$|S| = 5$

P(yes) = 3/5, P(no) = 2/5

Entropy(S) = 0.971

Step 2 : Entropy after Split

b)Based on Location

→ $|S_{MUM}| = 2$

P(yes) = 1, P(no) = 0

Entropy(S_{MUM}) = 0

→ $|S_{BLR}| = 2$

P(yes) = ½ , P(no) = ½

Entropy(S_{BLR}) = 1

→ $|S_{HYD}| = 1$

P(yes) = 0, P(no) = 1

Entropy(S_{HYD}) = 0

→ I(Location) = 2/5 * 1 = 0.4

→ Gain(S, Location) = 0.971 - 0.4 =

0.571

Construct Decision Tree

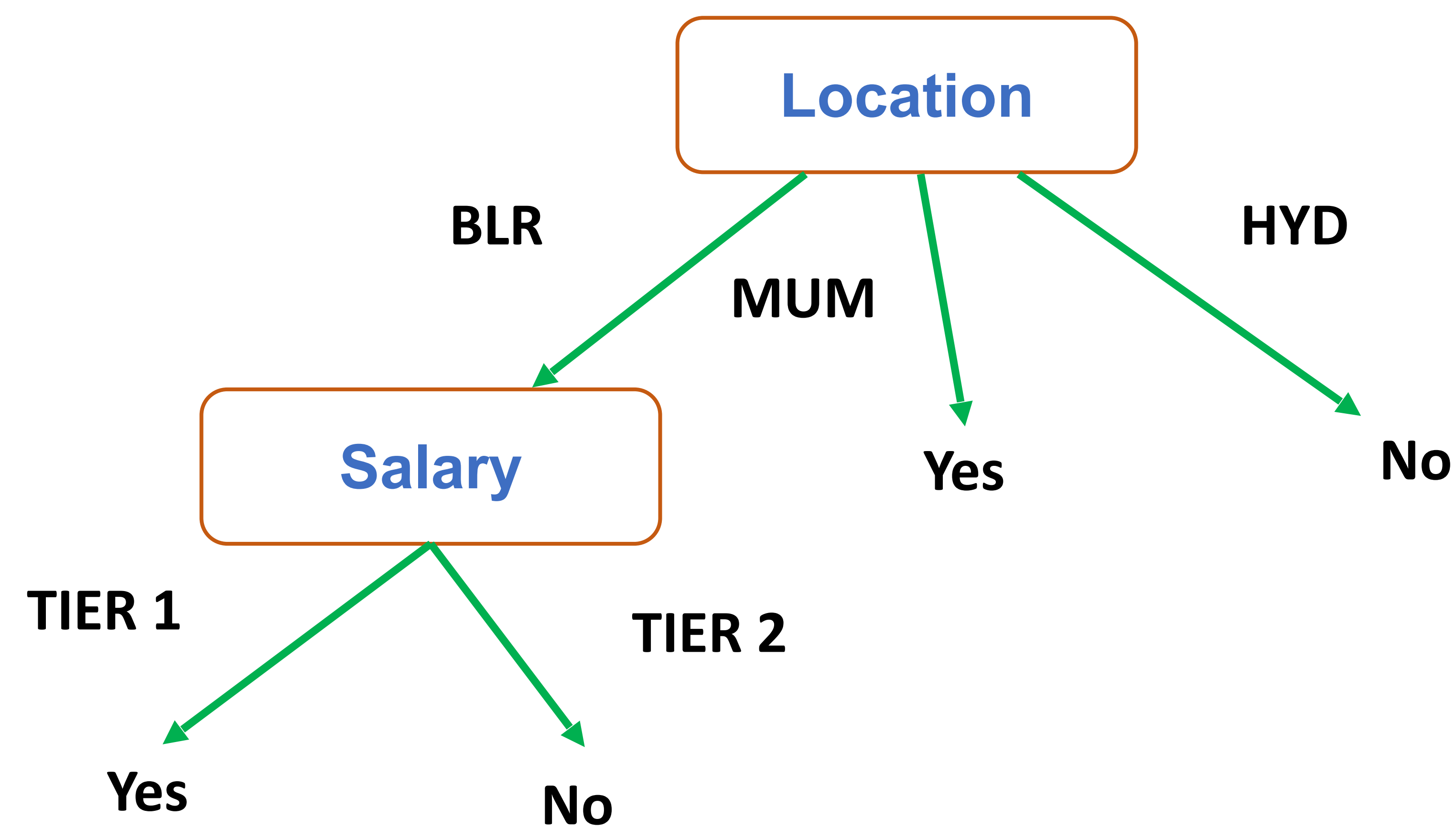
salary	Location	job acceptance
Tier1	MUM	YES
Tier 1	BLR	YES
Tier 2	BLR	NO
Tier 1	HYD	NO
Tier 2	MUM	YES

Decision Node - Max Gain

Gain(S, salary) = 0.0202

Gain(S, location) = 0.571

The root node : Location



M	N	O	Y
A	C	X	T
A	C	Z	E
A	D	X	T
A	D	Z	T
B	C	X	E
B	C	Z	F
B	D	X	F
B	D	Z	F

Example 3

M	N	O	Y
A	C	X	T
A	C	Z	E
A	D	X	T
A	D	Z	T
B	C	X	E
B	C	Z	F
B	D	X	F
B	D	Z	F

Step 1:
Compute entropy of the dataset:

No of points belonging to class T: 3
No of points belonging to class E: 2
No of points belonging to class F: 3
Total no. of points: 8

$$\text{Entropy(S)} = \frac{-3}{8}\log_2(\frac{3}{8}) - \frac{2}{8}\log_2(\frac{2}{8}) - \frac{3}{8}\log_2(\frac{3}{8})$$

$$\text{Entropy(S)} = \mathbf{1.56}$$

Note that if the target attribute can take on c possible values, the entropy can be as large as $\log_2 c$.

Machine Learning
Construct Decision Tree

Weather	Mood	Genre	Alone	Watch Movie
Sunny	Happy	Comedy	No	**Yes**
Sunny	Bored	Drama	Yes	**No**
Cloudy	Lazy	Thriller	Yes	**Yes**
Rainy	Sad	Romance	No	**Yes**
Rainy	Bored	Comedy	Yes	**No**
Rainy	Happy	Thriller	Yes	**No**
Cloudy	Happy	Romance	No	**Yes**
Sunny	Bored	Comedy	No	**No**
Sunny	Happy	Romance	Yes	**Yes**
Rainy	Lazy	Drama	No	**Yes**
Sunny	Lazy	Thriller	Yes	**Yes**
Cloudy	Bored	Comedy	Yes	**Yes**
Cloudy	Sad	Romance	No	**Yes**
Rainy	Sad	Thriller	Yes	**No**

Example 4

Entropy Calculation before Split

Weather	Mood	Genre	Alone	Watch Movie
Sunny	Happy	Comedy	No	**Yes**
Sunny	Bored	Drama	Yes	**No**
Cloudy	Lazy	Thriller	Yes	**Yes**
Rainy	Sad	Romance	No	**Yes**
Rainy	Bored	Comedy	Yes	**No**
Rainy	Happy	Thriller	Yes	**No**
Cloudy	Happy	Romance	No	**Yes**
Sunny	Bored	Comedy	No	**No**
Sunny	Happy	Romance	Yes	**Yes**
Rainy	Lazy	Drama	No	**Yes**
Sunny	Lazy	Thriller	Yes	**Yes**
Cloudy	Bored	Comedy	Yes	**Yes**
Cloudy	Sad	Romance	No	**Yes**
Rainy	Sad	Thriller	Yes	**No**

ENTROPY(S) - Calculating for entire Set S

|S| = 14 (training instances provided)

P(watch movie = yes) = p1 = 9/14 = 0.643

P(Play tennis = no) = p2 = 5/14 = 0.357

$$Entropy(S) \equiv \sum_{i=1}^{c} -p_i \log_2 p_i$$

$$= -(p1 \log_2 p1 + p2 \log_2 p2)$$

$$= 0.94$$

Calculating Entropy for a Subset

- We will now see, What happens if we split based on:
  - Weather
  - Mood
  - Genre
  - Alone

- First, we will see what will happen If we split based on Weather.
  - Weather has 3 values: three branches = sunny, cloudy, and rainy.
  - Calculate the entropy for each of these subsets.
  - Weighted sum of the entropy's of the three subsets which becomes the "Entropy after the split".
  - We then calculate the gain(weather) = Entropy before split - Entropy after split

- We repeat this process for each attribute.

- The attribute with the maximum gain is picked as the decision node.

Machine Learning
Calculating Entropy for a Subset

Weather	Mood	Genre	Alone	Watch Movie
Cloudy	Lazy	Thriller	Yes	Yes
Cloudy	Happy	Romance	No	Yes
Cloudy	Bored	Comedy	Yes	Yes
Cloudy	Sad	Romance	No	Yes
Rainy	Sad	Romance	No	Yes
Rainy	Bored	Comedy	Yes	No
Rainy	Happy	Thriller	Yes	No
Rainy	Lazy	Drama	No	Yes
Rainy	Sad	Thriller	Yes	No
Sunny	Happy	Comedy	No	Yes
Sunny	Bored	Drama	Yes	No
Sunny	Bored	Comedy	No	No
Sunny	Happy	Romance	Yes	Yes
Sunny	Lazy	Thriller	Yes	Yes

Entropy after Split based on Weather and Gain Calculation

$|S| = 14$, Entropy(S) = 0.94

- Splitting based on weather - there will be 3 subsets
  $|S_{sunny}| = 5$, $|S_{cloudy}| = 4$ and $|S_{rainy}| = 5$

- Entropy of these subsets:
  - $E(S_{sunny}) = -(\frac{2}{5} \log_2 \frac{2}{5} + \frac{3}{5} \log_2 \frac{3}{5}) = 0.971$
  - $E(S_{cloudy}) = -(1 \log_2 1 + 0 \log_2 0) = 0$
  - $E(S_{rainy}) = -(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5}) = 0.971$

- Weighted avg of the entropies of the subsets:
  I(weather) = $\sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$ = 5/14 * 0.971 + 4/14 * 0 + 5/14 * 0.971
  = 0.693

- Gain(S, outlook) = Entropy(S) - I(outlook) = 0.94 - 0.693 = **0.247**

Picking an Attribute as Decision Node

- We do a similar calculation (by splitting based on the other attributes) to calculate the gain

- We can summarize the calculations as:

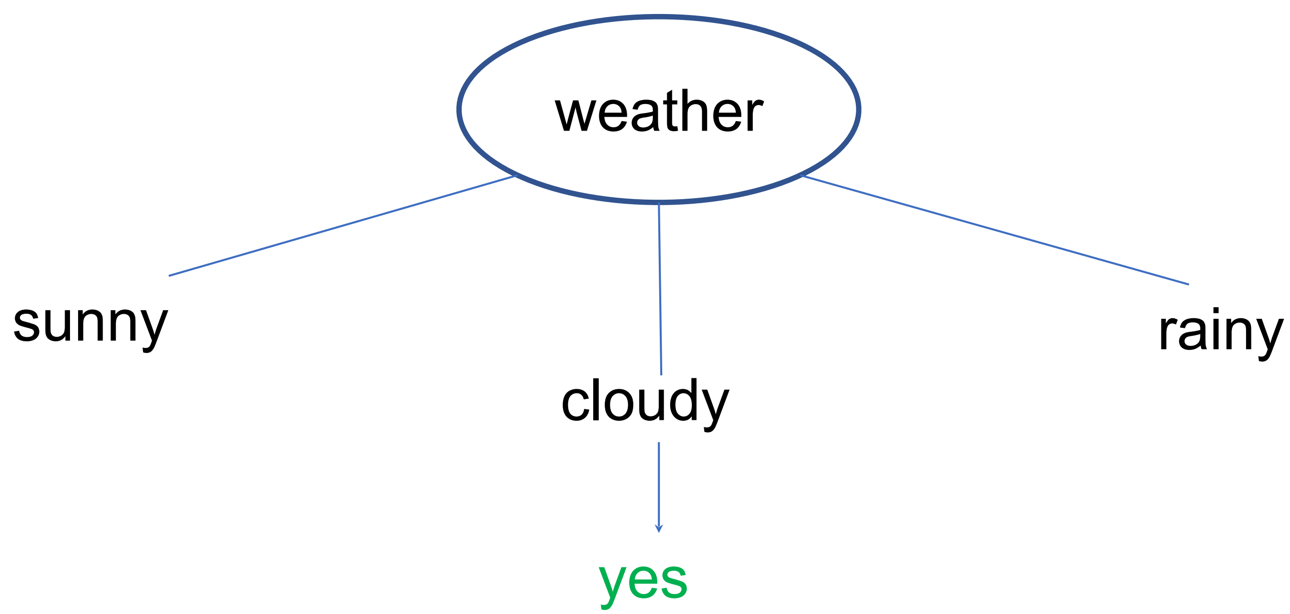
property	value
Entropy(S)	0.94
G(S, weather)	0.247
G(S, mood)	…
G(S, genre)	…
G(S, alone)	…

- We pick the attribute with maximum Gain as the decision node.

Arriving at a Decision Node - Attribute with highest Gain

- We choose our root node as weather and split according to its three values.
- Entropy of overcast was 0 which means it is single valued and hence we can come to a conclusion from overcast .

property	value
Entropy(S)	0.94
G(S, weather)	0.247
G(S, mood)	…
G(S, genre)	…
G(S, alone)	…

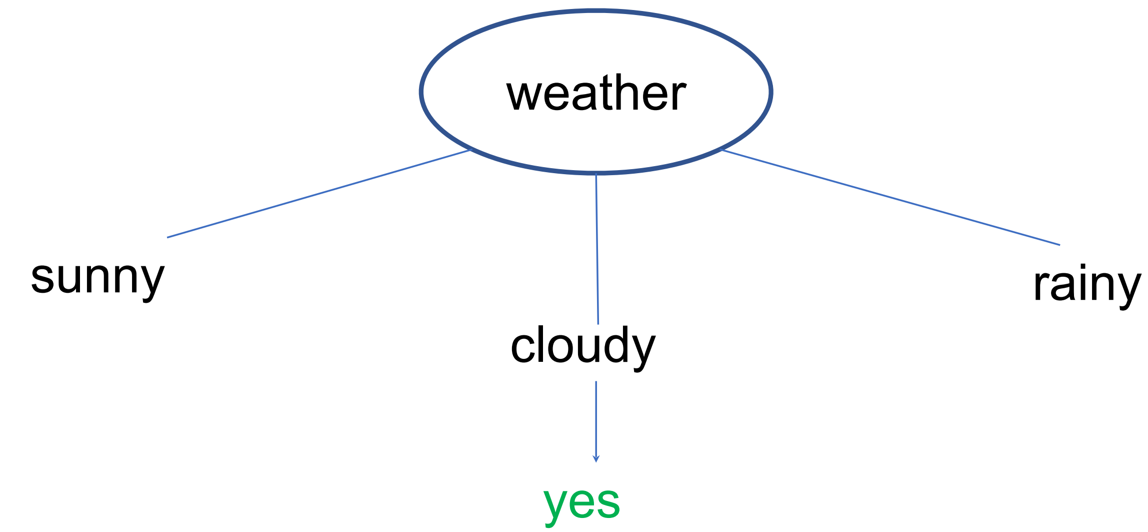


$E(S_{sunny})$	0.971
$E(S_{cloudy})$	0
$E(S_{rainy})$	0.971

Repeat the process to pick the descendant node for each branch

With Outlook = Sunny, data would look like this:

Weather	Mood	Genre	Alone	Watch Movie
Sunny	Happy	Comedy	No	Yes
Sunny	Bored	Drama	Yes	No
Sunny	Bored	Comedy	No	No
Sunny	Happy	Romance	Yes	Yes
Sunny	Lazy	Thriller	Yes	Yes



$E(S_{sunny})$	0.971
$E(S_{cloudy})$	0
$E(S_{rainy})$	0.971

Entropy after Split based on Mood and Gain Calculation

$|S_{sunny}| = 5$, $Entropy(S_{sunny}) = 0.971$

- Splitting based on Mood - there will be 3 subsets
  $|S_{happy}| = 2$, $|S_{bored}| = 2$ and $|S_{lazy}| = 1$

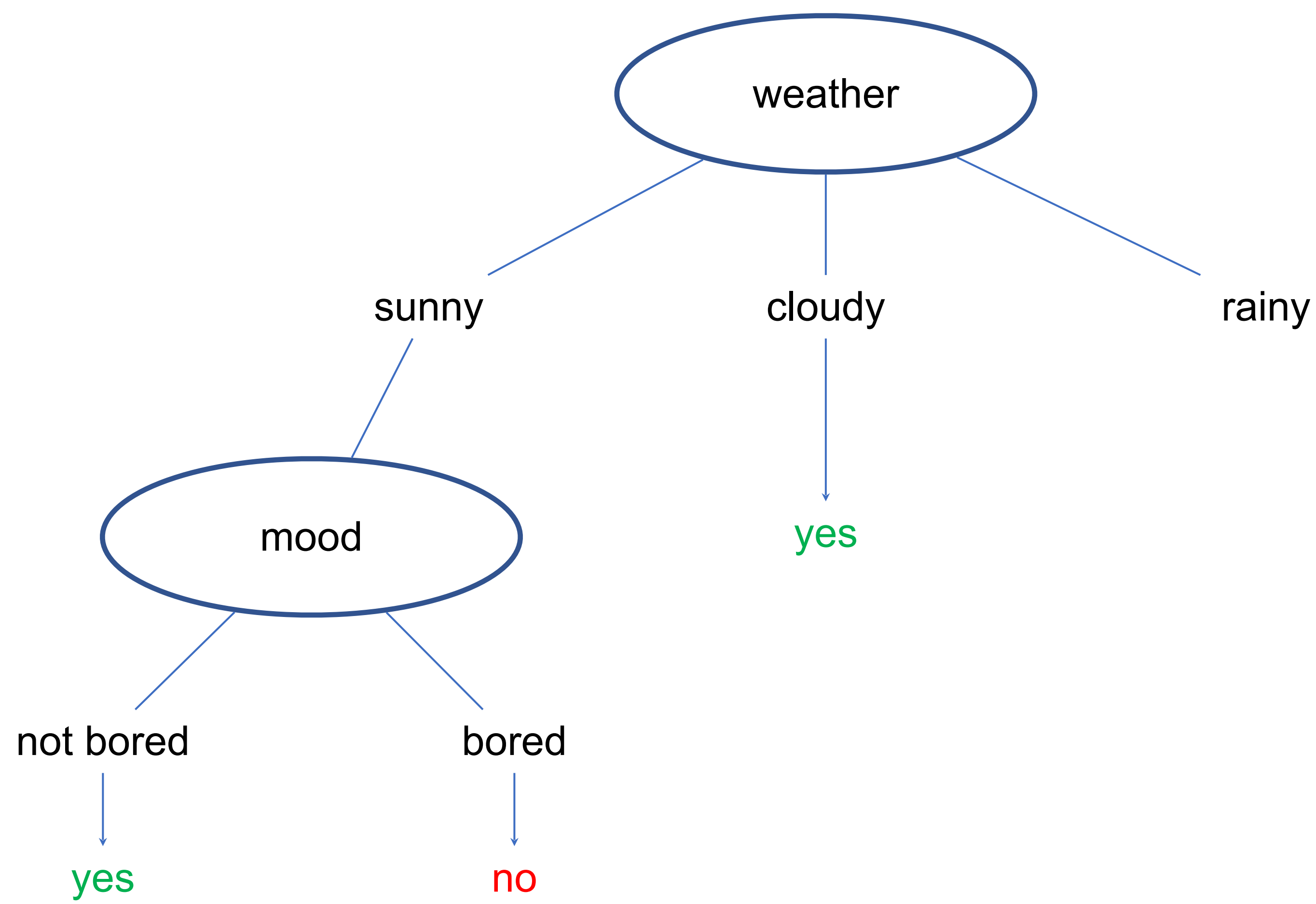
- Entropy of these subsets:
  - $E(S_{happy}) = -(1 \log_2 1 + 0 \log_2 0) = 0$
  - $E(S_{bored}) = -(0 \log_2 0 + 1 \log_2 1) = 0$
  - $E(S_{lazy}) = -(1 \log_2 1 + 0 \log_2 0) = 0$

- Avg Information of Mood: weighted avg of the entropies of the subsets:
  $I(Mood) = \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v) = 0$

- $Gain(S_{sunny}, Mood) = Entropy(S_{sunny}) - I(Mood) = 0.971 - 0 =$ **0.971**

Arriving at a Decision node

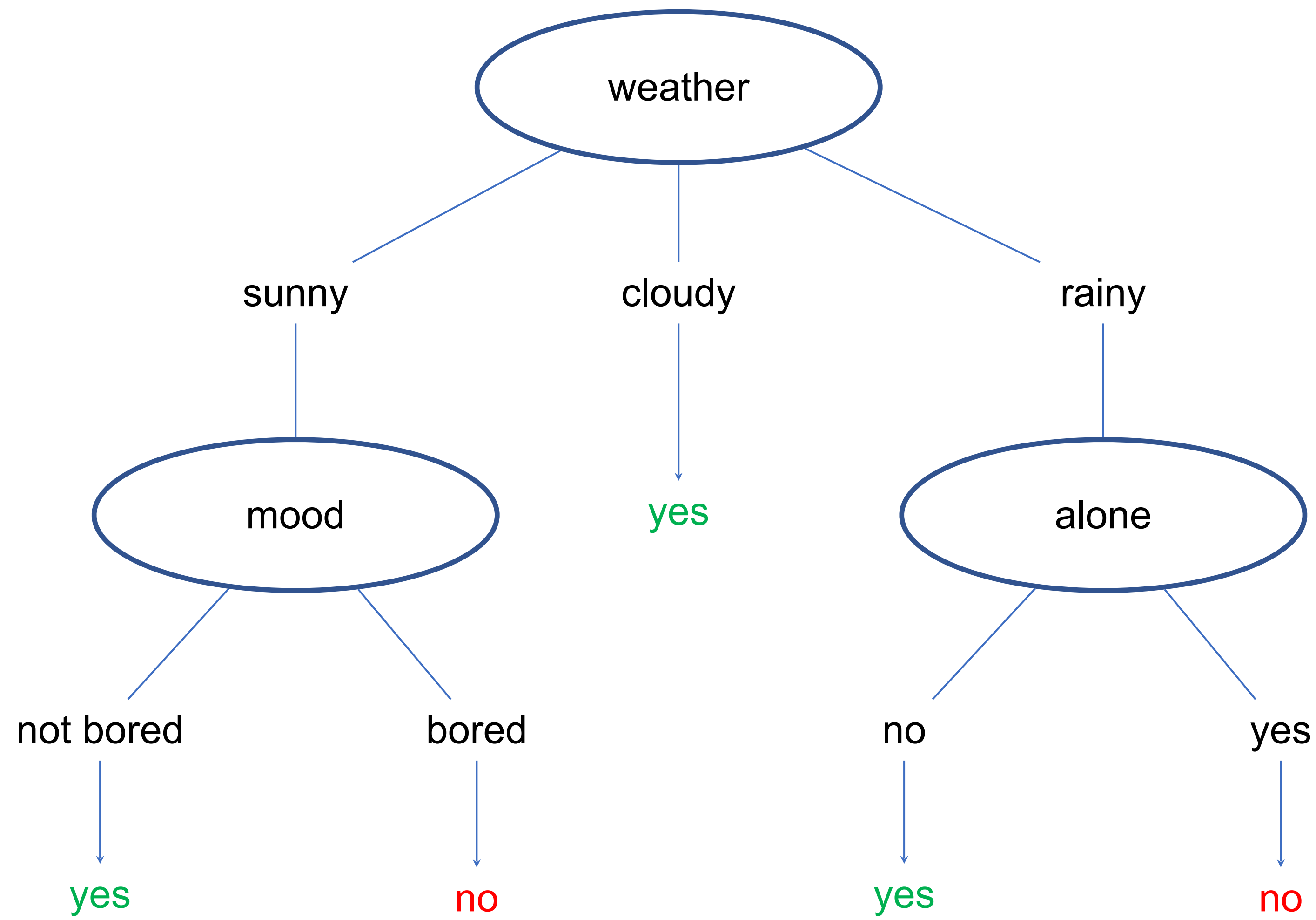
- Mood has the highest gain hence we choose mood at second level after weather as sunny
- Remember that both value of all values of mood have entropy of 0 which means we have reached the leaf node with this.



property	value
Entropy(S_{sunny})	0.971
G(S_{sunny}, genre)	…
G(S_{sunny}, mood)	0.971
G(S_{sunny}, alone)	…

Arriving at a Decision node

- Doing the same procedure for the table with weather=rainy we expand the rainy leading node to get the following decision tree.



Machine Learning

Dr. Shylaja S S

Department of Computer Science & Engineering

shylaja.sharath@pes.edu